Independent model - experimental results

CS-433 Machine Learning, EPFL | Prepared from Ziyad's NumPy pipeline and documented experiments | 27 October 2026

1. Objective and model

Predict the binary survey label _MICHD (reported coronary heart disease / myocardial infarction). Dataset: 328,135 training cases, 109,379 competition cases, 321 features; 28,975 positive training labels (8.83%). A model predicting only the negative class reaches 91.17% accuracy but F1 = 0.

Model: L2-regularized logistic regression optimized with mini-batch Adam (batch size 8,192; learning rate 0.03; L2 = 0.0002). Only NumPy and Python standard libraries are used for modeling. Features are fitted using the training portion of each split.

2. Preprocessing and experimental design

Drop bookkeeping features and variables with insufficient observations; one-hot encode categorical answers; handle known special codes for selected numerical variables; impute, clip and scale numerical features, and add missingness indicators. The initial 70%-training split yielded 691 engineered features including the intercept.

Local tests use stratified 70/15/15 train/tune/evaluation splits. Tuning chooses stopping epoch and the threshold that maximizes positive-class F1; evaluation uses the locked threshold. Seeds: 42, 123, 2026.

3. Ablation: seven handcrafted interactions

Feature set	Seed 42	Seed 123	Seed 2026	Mean ± SD
Clinical features	0.4311	—	—	1 split only
Encoded, no interactions	0.4339	0.4347	0.4288	0.43245 ± 0.00320
Encoded + 7 interactions	0.4342	0.4334	0.4296	0.43240 ± 0.00243

The interactions have no stable improvement: mean paired effect (with minus without) ≈ −0.00005 F1. The simpler encoded model is retained. These comparisons are exploratory, not a pristine final unbiased estimate after model selection.

4. Seed-42 held-out diagnostics (no interactions)

Metric	Result	Metric	Result
Train / tune / evaluation	229,695 / 49,220 / 49,220	Encoded dimensions	691
Selected epoch	6	Tuning threshold	0.230
Evaluation F1	0.433878	Evaluation accuracy	0.876148
Evaluation precision	0.363750	Evaluation recall	0.537506
True positives / false positives	2,336 / 4,086	False negatives / true negatives	2,010 / 40,788

At the default threshold 0.5, held-out F1 is 0.2525; at the tuned threshold 0.230, it is 0.4339. The threshold curve is plotted using tuning data; the evaluation set is scored only at the fixed threshold.

5. AIcrowd result and reproducibility

AIcrowd submission #338016 (27 October 2026): F1 = 0.436; accuracy = 0.870. This is a competition score, not the same as the local evaluation. The model describes association with a self-reported historical diagnosis, not a prospective clinical risk assessment.

All recorded ablation results are in documentation/ziyad_results.json. Recreate the figures with python documentation/ziyad_figures.py or rerun the full ablation using the same script with --recompute-ablation. Keep the exact CSV uploaded to AIcrowd and verify that run.py reproduces it byte-for-byte (e.g. SHA-256), before treating it as the final group submission.

Note: these experimental results belong to Ziyad's model. Ismail's Newton-based model uses a different validation protocol; local scores must not be presented as directly comparable.